

Israel Souza Ferreira

Data Scientist | Machine Learning Engineer

Fortaleza, Brazil | Remote | +55 85 98433-2790 | israel.souza.dev@gmail.com
[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [Lattes](#)

PROFILE

Data scientist and machine learning engineer experienced in **applied research and end-to-end delivery**: experimentation, computer vision, APIs, data, and CPU-only deployment. I support decisions with baselines, ablations, benchmarks, and automated testing.

PROFESSIONAL EXPERIENCE

Tieta Artificial Intelligence

R&D Researcher — Research and Innovation Training Program

Remote, Brazil

Jul 2026 - Present

- Developing an **OCR and document-segmentation pipeline** that turns scanned exams into structured questions for educational analysis.

Tieta Artificial Intelligence

Data Scientist / R&D Researcher — CNPq RHAЕ Research Fellow

Remote, Brazil

Apr 2025 - Jun 2026

- Modeled psychometric parameters with **Item Response Theory** and compared approaches through paired evaluations, ablations, correlation with references, and inference cost.
- Built signal-extraction pipelines with LLMs orchestrated by **DSPy** and evaluated multitask regression architectures over textual and visual data through reproducible experiments.

Independent Consulting — Educational Institution

Full Stack Developer and Machine Learning Engineer

Remote, Brazil

2026 - Present

- Built the **Face Clock Evoluir**, a contracted facial-attendance system in final acceptance testing. Owned models, backend, database, security, and deployment; the current delivery has **574 automated tests**.
- Put 1:N recognition into CPU-only operation with YuNet + SFace, ONNX Runtime, and PostgreSQL/pgvector: **74 ms p95** with one identification. Open-set calibration reduced accepted unknowns from **45.4% to 0.2%** on public reference data.
- Experiments reduced **5 to 3 frames** without measured loss and showed **0.076 ms** exact search over 1,000 vectors; kept transactional pgvector search instead of adding FAISS without a demonstrated gain.
- Compared five approaches for medical-document triage; EfficientNet + Logistic Regression reached **87.5% accuracy** with seven fewer false positives at the same recall. For an 8 GB, GPU-less server, reduced images from **10.59 to 1.11 GB**, with no classifier decisions changed in the migration to ONNX Runtime.

Federal University of Ceará — Quixadá Campus

Teaching Assistant for Algorithms, Calculus, and Pre-Calculus

On-site, Brazil

2022 - 2024

- Supported classes in data structures, algorithms, complexity, and Calculus, translating quantitative material across skill levels.

SELECTED PROJECTS

- Triple Roman Domination in Graphs (undergraduate thesis)**: first GA- and ACO-based metaheuristics reported for the problem, plus a correction to the published ILP formulation; evaluated over **362 graphs**.
- Brazilian Emergency Aid — Data Engineering**: chunked binary COPY ingestion over **31.6 GB** of public data; instrumentation linked memory growth to retained deduplication IDs (**r = 0.99**).

TECHNICAL SKILLS

Languages and querying: Python, SQL, C++, and Bash.

ML, vision, and experimentation: PyTorch, Scikit-learn, OpenCV, YuNet, SFace, ONNX Runtime, DSPy, Optuna, cross-validation, calibration, ablation, hypothesis testing, and vector search.

Data and production: PostgreSQL, pgvector, FAISS, Pandas, NumPy, SciPy, FastAPI, SQLAlchemy, Alembic, Docker, GitHub Actions, pytest, MLflow, and Streamlit.

EDUCATION

Federal University of Ceará — Quixadá Campus

B.Sc. in Computer Science

Brazil

Jul 2026

LANGUAGES

Portuguese: Native | **English**: Fluent technical reading of research papers and documentation